Driving the Future of Rail

Efficient train booking and passenger management using Data Structures

DS PROJECT

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Project Overview

- The Railway Reservation System is a C-based application designed to simulate the process of railway ticket reservation.
- It allows passengers to:
  - View available trains
  - Search for trains
  - Book tickets
  - Cancel tickets
  - Join a waiting list when seats are unavailable
  - Search for passenger bookings
  - View all current bookings
- The system uses different data structures to efficiently manage trains, passengers, bookings, cancellations, and waiting lists.
- File handling is used to save important records so that data can be retained.

Our project simulates a real-world railway reservation system using C programming and different data structures

The Problem

The Challenges

Traditional manual reservation systems can become difficult to manage when there are:

- Large numbers of passengers
- Limited seats
- Multiple trains
- Ticket cancellations
- Waiting-list passengers
- Frequent searching of passenger records

Our Solution

The Approach

Develop a computerized railway reservation system that can:

- Manage train information
- Manage available seats
- Store passenger information
- Automate ticket booking
- Handle cancellations
- Automatically process waiting-list passengers
- Search and display records efficiently
- Store records using files

Objectives of the Project

Innovate. Connect. Sustain.

- Train Management
  Store train number, name, route and seat availability.
- Train Searching
  Search trains using train number or train name.
- Ticket Booking
  Accept passenger details and allocate available seats.
- Ticket Cancellation
  Cancel tickets and release occupied seats.
- Waiting List Management
  Add passengers to a queue when all seats are occupied.
- Automatic Waiting List Processing
  Assign a newly available seat to the first waiting passenger.
- Passenger Searching
  Search bookings using ticket number or passenger name.
- Data Storage
  Save records using file handling.

Main Features

1. Display Trains
   - Displays train number
   - Train name
   - Source and destination
   - Available seats

2. Search Train
   - Search using train number
   - Search using train name

3. Book Ticket
   - Enter passenger details
   - Select train
   - Allocate seat
   - Generate ticket number

4. Cancel Ticket
   - Find ticket
   - Cancel booking
   - Release seat

5. Waiting List
   - If seats are full, passenger joins waiting queue.

6. Process Waiting List
   - Cancelled seat is automatically assigned to the first waiting passenger.

7. Search Passenger
   - Search using name or ticket number.

8. Display Bookings
   - Display all confirmed passengers.

System Workflow

Our Project Offers

Start
↓
Display Main Menu
↓
Choose Operation
↙ ↓ ↓ ↓ ↘
Display Trains | Search | Book | Cancel | Search Passenger
↓
For Booking:
Enter Passenger Details
↓
Check Seat Availability
↓
Seat Available?
YES → Allocate Seat → Generate Ticket → Save Record
NO → Add Passenger to Waiting Queue
For Cancellation:
Enter Ticket Number
↓
Find Booking
↓
Cancel Ticket
↓
Release Seat
↓
Check Waiting Queue
↓
Assign Seat to First Waiting Passenger
↓
Update Records
↓
End

Data Structures Used

DATA STRUCTURES USED IN THE PROJECT

Data Structure	Purpose
Array	Store train and seat information
Linked List	Store passenger/booking records
Queue	Manage waiting-list passengers
Stack	Maintain cancellation/history information
Searching	Find trains and passengers
Sorting	Arrange records

Future Scope

Graphical User Interface

Create a GUI instead of a console interface.

Online Database

Use MySQL or another database to store records.

User Authentication

Add:

- Login
- Password
- Admin account
- Passenger account

Future Scope

Online Payment

Integrate payment gateways for ticket booking.

Real-Time Train Information

Show:

- Train status
- Delays
- Platform number
- Live seat availability

Mobile Application

Convert the system into an Android/iOS application.

Conclusion

- The Railway Reservation System demonstrates how C programming and data structures can be applied to a real-world problem.
- Arrays are used for train and seat information.
- Linked lists manage dynamic passenger records.
- Queues handle waiting-list passengers using FIFO.
- Stacks maintain cancellation/history information.
- Searching and sorting make record management easier.
- File handling allows important information to be stored.

The project provides a practical understanding of how multiple data structures can work together in one application.

The project demonstrates how data structures can make reservation management organized, efficient, and systematic.

Thank you so much

We aim to leverage technology and data structures to create an efficient, reliable, and user-friendly railway reservation system, contributing to a smarter and more organized future of railway management